

EXERCISE (O-1)

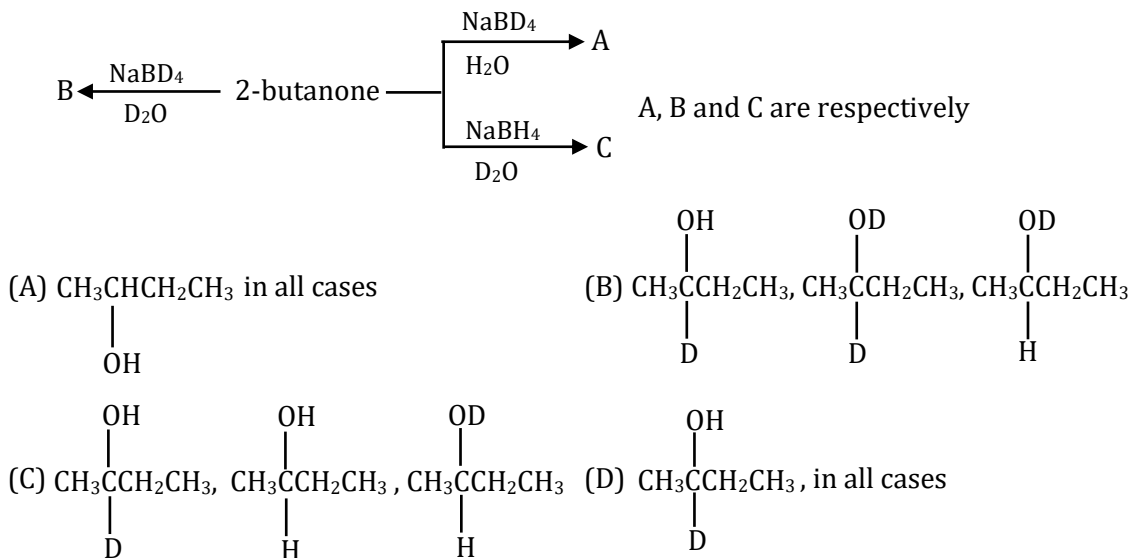
SINGLE CORRECT TYPE QUESTIONS

Metal hydride reduction (LiAlH_4 ; NaBH_4 ; DIBAL-H; B_2H_6 ; LiBH_4 ; NaBH_3CN)

1. The product of the reaction $\text{Ph}_2\text{C} = \text{O} \xrightarrow[\text{H}_3\text{O}^+]{\text{LiAlD}_4}$ is
 (A) $\text{Ph}_2\text{CD}(\text{OH})$ (B) $\text{Ph}_2\text{CH}(\text{OD})$ (C) $\text{Ph}_2\text{CD}(\text{OD})$ (D) None

CRD086

2. Consider reduction of 2-butanone.



CRD087

Catalytic hydrogenation (Except alkene and alkyne)

3. The relative rates of hydrogenation is in the order of :
 (A) $\text{CH}_2 = \text{CH}_2 > \text{RCH} = \text{CH}_2 > \text{RCH} = \text{CHR} > \text{R}_2\text{C} = \text{CHR}$
 (B) $\text{R}_2\text{C} = \text{CHR} > \text{RCH} = \text{CHR} > \text{RCH} = \text{CH}_2 > \text{CH}_2 = \text{CH}_2$
 (C) $\text{RCH} = \text{CHR} > \text{R}_2\text{C} = \text{CHR} > \text{RCH} = \text{CH}_2 > \text{CH}_2 = \text{CH}_2$
 (D) $\text{R}_2\text{C} = \text{CHR} > \text{CH}_2 = \text{CH}_2 > \text{RCH} = \text{CHR} > \text{RCH} = \text{CH}_2$

CRD053

4.
$$\begin{array}{c} \text{H}_3\text{C} \quad \text{C}_2\text{H}_5 \\ \quad \diagdown \quad \diagup \\ \quad \text{C} \\ \quad \parallel \\ \quad \text{C} \\ \diagup \quad \diagdown \\ \text{C}_2\text{H}_5 \quad \text{CH}_3 \end{array} \xrightarrow{\text{H}_2/\text{Pt}} \text{x, x is}$$

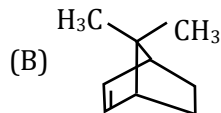
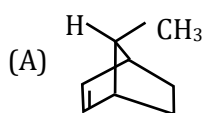
 (A) Only (R,R) (B) Only (S,S) (C) Meso compound (D) Racemic mixture

CRD089

5. Which of the following reactions involves homogeneous reduction
 (A) $\text{CH}_2 = \text{CH}_2 \xrightarrow{\text{H}_2/\text{Ni}} \text{CH}_3 - \text{CH}_3$
 (B) $\text{CH}_3 - \text{C} \equiv \text{C} - \text{CH}_3 \xrightarrow{\text{Wilkinson's catalyst}} \text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$
 (C) $\text{CH}_3\text{COCl} \xrightarrow{\text{H}_2/\text{Pd}/\text{CaCO}_3/\text{quinoline}} \text{CH}_3\text{CHO}$
 (D) $\text{CH}_3\text{C} \equiv \text{CH} \xrightarrow{\text{H}_2/\text{Pd}/\text{CaCO}_3/\text{quinoline}} \text{CH}_3\text{CH} = \text{CH}_2$

CRD088

6. Which undergoes catalytic hydrogenation much faster ?

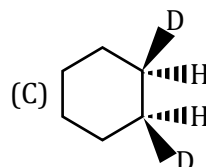
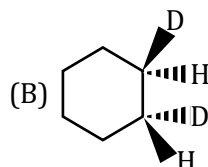
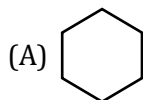


(C) equally

(D) cannot be predicted

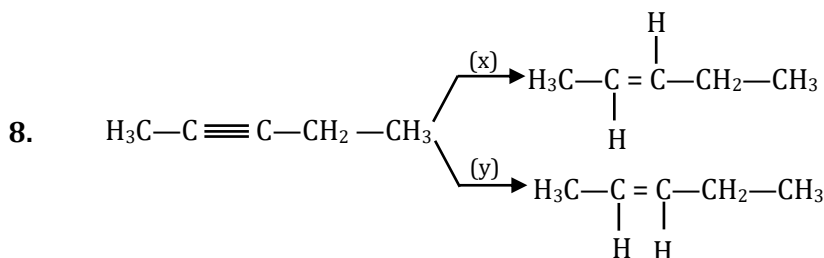
CRD090

7. $\xrightarrow[\text{H}_2\text{O}_2]{\text{NH}_2\text{-NH}_2}$ Product:



(D) Both (B) and (C)

CRD003



In the above reaction the using reagents X and Y are

(A) Na/liq. NH_3 for X

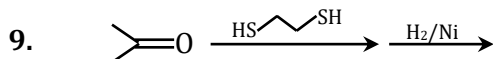
(B) H_2 , Pd/ BaSO_4 for Y

(C) $\text{BH}_3\text{-THF} + \text{CH}_3\text{COOH}$ for Y

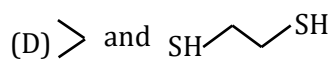
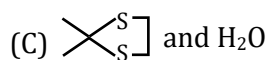
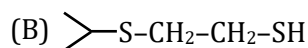
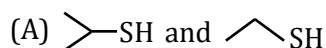
(D) All of these are correct

CRD091

Mozingo Reduction ($\text{HSCH}_2\text{CH}_2\text{SH} + \text{Dry HCl} + \text{Raney Ni}$)



The end products of the reaction are :



CRD001

Stephen's reduction (SnCl_2/HCl)

10. Stephen reduction (SnCl_2/HCl) converts cyanides to

(A) Aldehydes

(B) Ketones

(C) Amines

(D) Acids

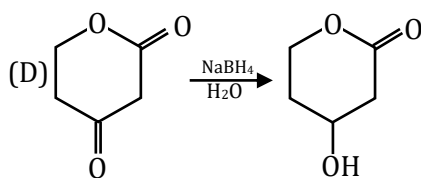
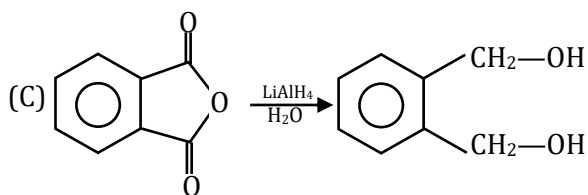
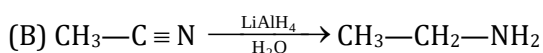
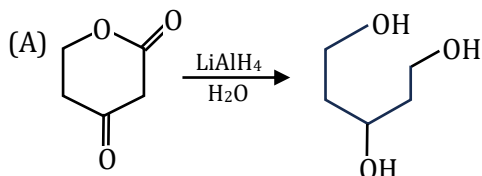
CRD056

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

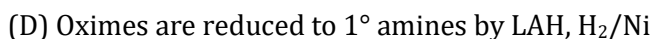
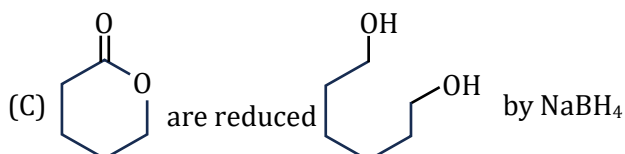
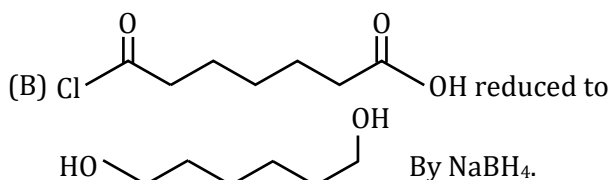
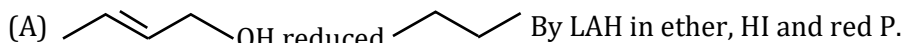
Metal hydride reduction (LiAlH_4 ; NaBH_4 ; DIBAL-H; B_2H_6 ; LiBH_4 ; NaBH_3CN)

1. Which of the following reactions is/ are correct?

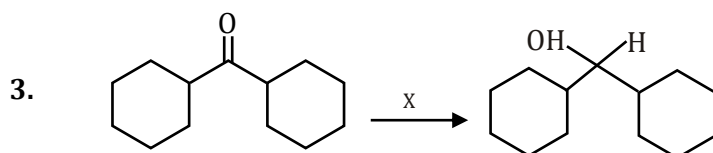


CRD092

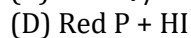
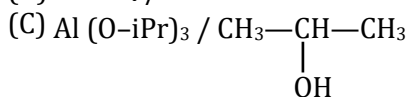
2. Which of the following is (are) not correct?



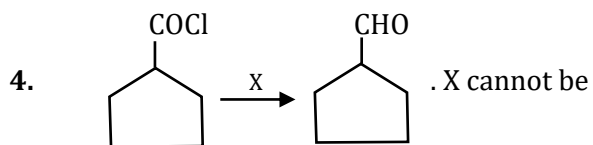
CRD129



X is / are

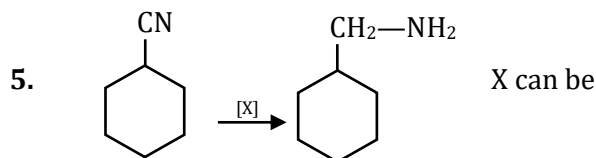


CRD094

Catalytic hydrogenation (Except alkene and alkyne)

- (A) $\text{NaBH}_4 / \text{EtOH}$ (B) $\text{LiAlH}_4 / \text{THF}$ (C) Na / EtOH (D) $\text{H}_2 / \text{Pd} - \text{BaSO}_4$

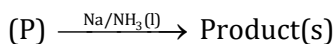
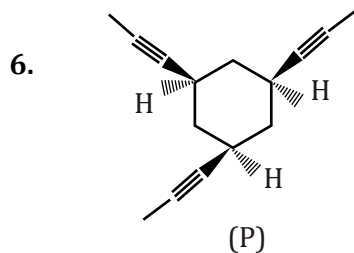
CRD095



- (A) H_2 / Ni (B) LiAlH_4 (C) NaBH_4 (D) $\text{Na} + \text{EtOH}$

CRD096

COMPREHENSION TYPE QUESTIONS

Catalytic hydrogenation (Except alkene and alkyne)

The products (s) has/ have degree of unsaturation and number of isomeric product (s) formed are, respectively

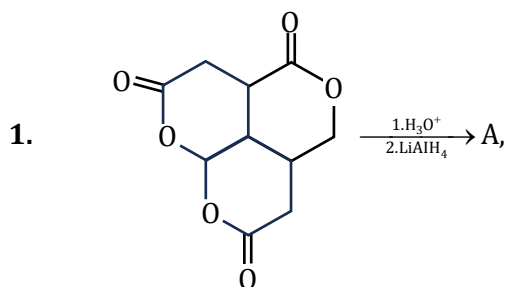
- (A) 1,4 (B) 4,2 (C) 4,1 (D) 2,4

CRD098

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

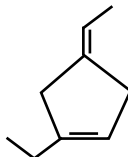
Metal hydride reduction (LiAlH_4 ; NaBH_4 ; DIBAL-H; B_2H_6 ; LiBH_4 ; NaBH_3CN)

How many $-\text{OH}$ group in the product (A) is

CRD101

Catalytic hydrogenation (Except alkene and alkyne)

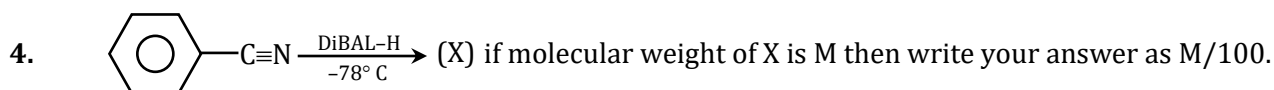
2. On catalytic reduction with H_2/Pt how many possible alkenes will give n-butane ?
3. If the following compound is treated with Pd in excess of H_2 gas, how many stereoisomers of the product will be obtained ?



CRD011

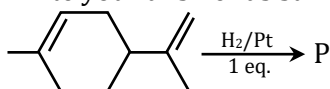
Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.



CRD130

5. For the product (P) given below, x is degree of unsaturation and y is total number of products then write your answer as sum of x + y is.

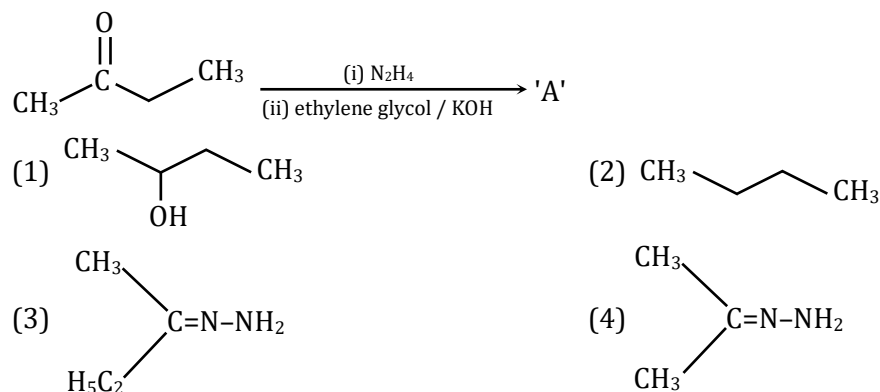


CRD131

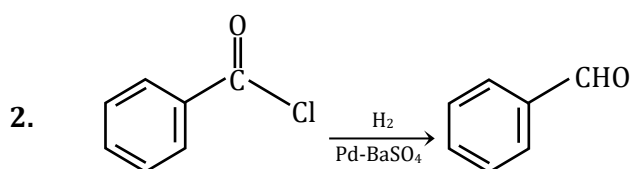
EXERCISE - JEE (Main) PYQ

1. Identify 'A' in the following reaction :

[JEE (Main) 2024]



CRD132



This reduction reaction is known as:

[JEE (Main) 2024]

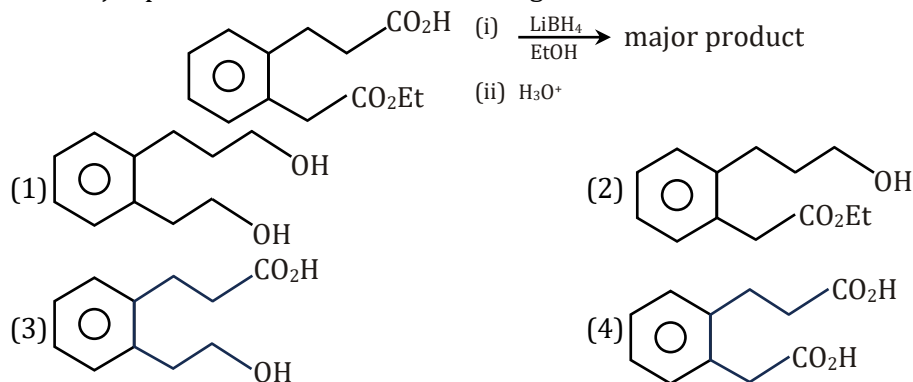
- (1) Rosenmund reduction
(3) Stephen reduction

- (2) Wolff-Kishner reduction
(4) Etard reduction

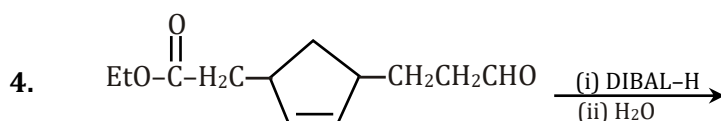
CRD133

3. The major product formed in the following reaction is:

[JEE (Main) 2023]

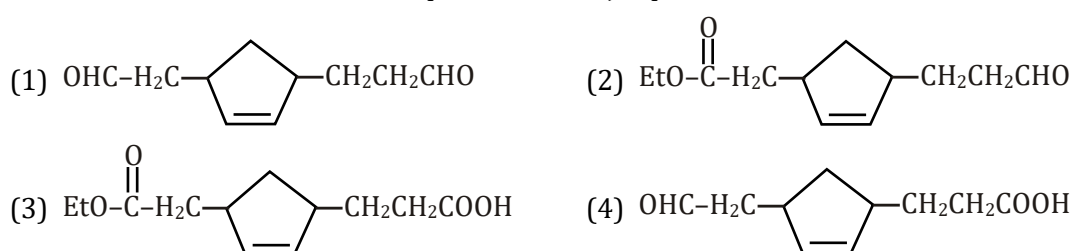


CRD111



[JEE (Main) 2022]

Consider the above reaction and predict the major product.



CRD049

5. **Statement I :** Sodium hydride can be used as an oxidising agent. [JEE (Main) 2021]

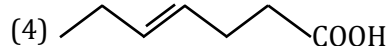
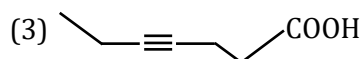
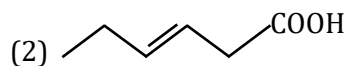
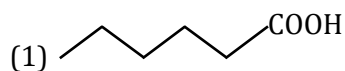
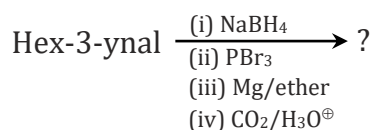
Statement II : The lone pair of electrons on nitrogen in pyridine makes it basic.

Choose the CORRECT answer from the options given below :

- (1) Both statement I and statement II are false
- (2) Statement I is true, but statement II is false
- (3) Statement I is false, but statement II is true
- (4) Both statement I and statement II are true

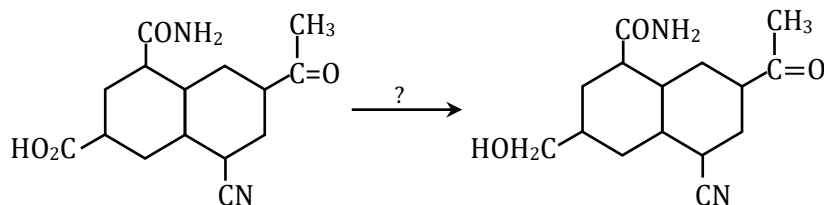
CRD045

6. What is the product of following reaction ? [JEE (Main) 2020]



CRD043

7. The most suitable reagent for the given conversion is : [JEE (Main) 2020]

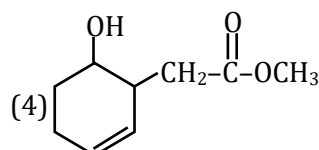
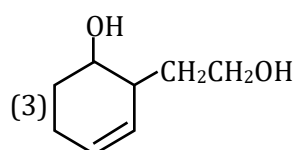
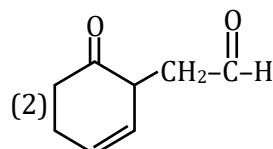
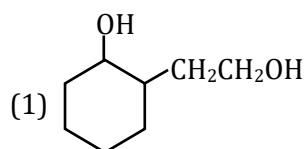
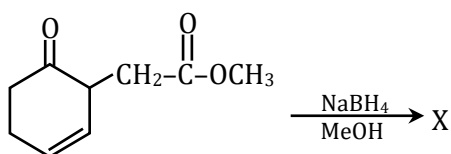


- (1) LiAlH_4
- (3) H_2/Pd

- (2) NaBH_4
- (4) B_2H_6

CRD044

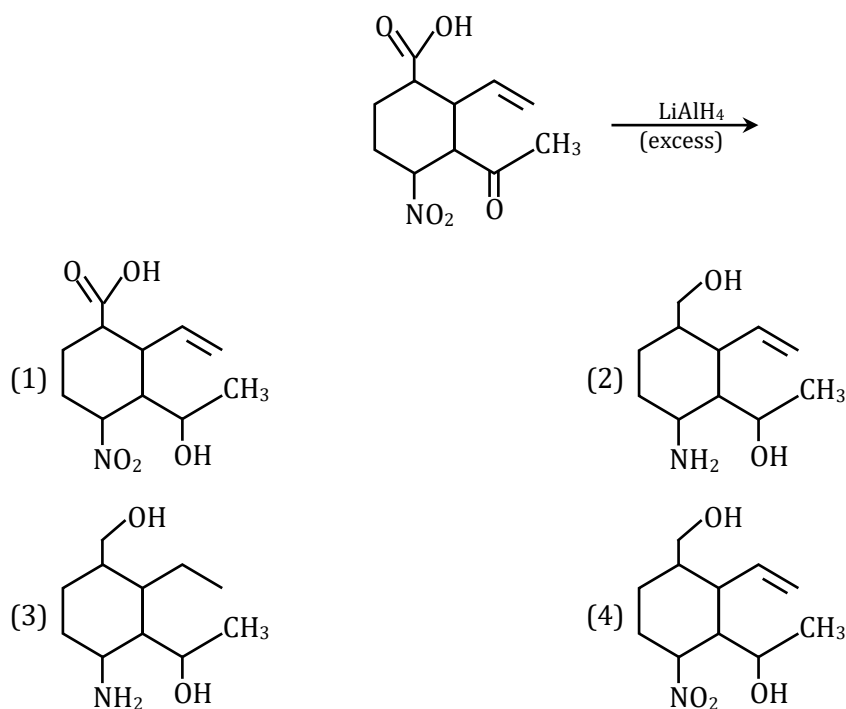
8. The major product 'X' formed in the following reaction is : [JEE (Main) 2019]



CRD037

9. The major product obtained in the following reaction is :-

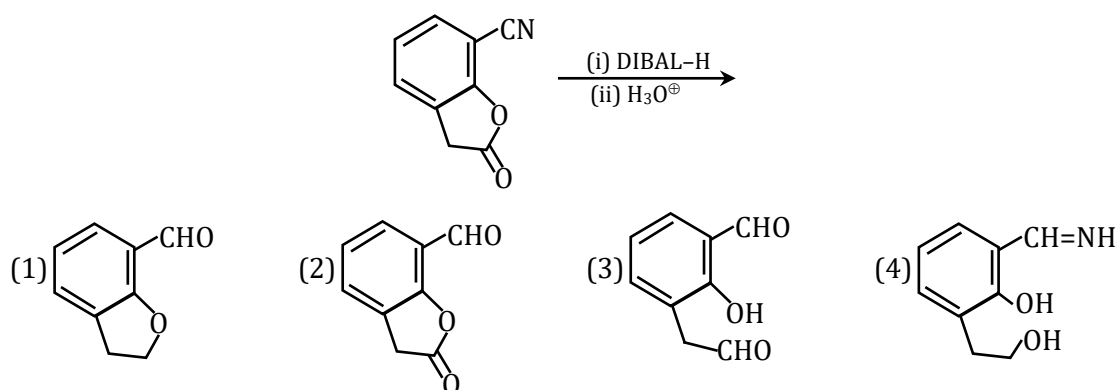
[JEE (Main) 2019]



CRD038

10. The major product of the following reaction is:

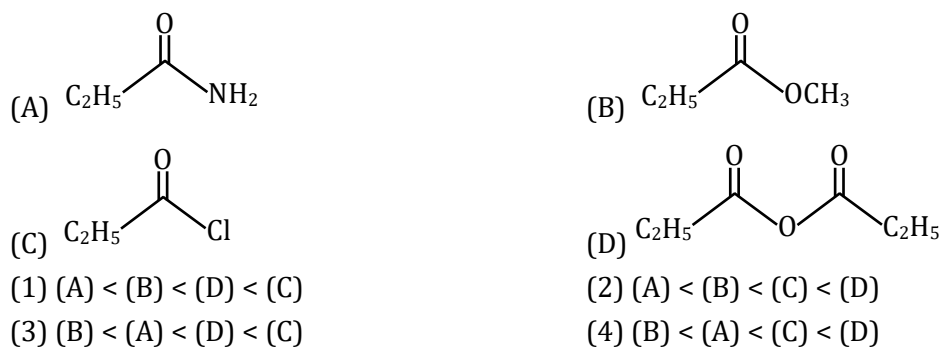
[JEE (Main) 2019]



CRD039

11. The increasing order of the reactivity of the following with LiAlH_4 is :

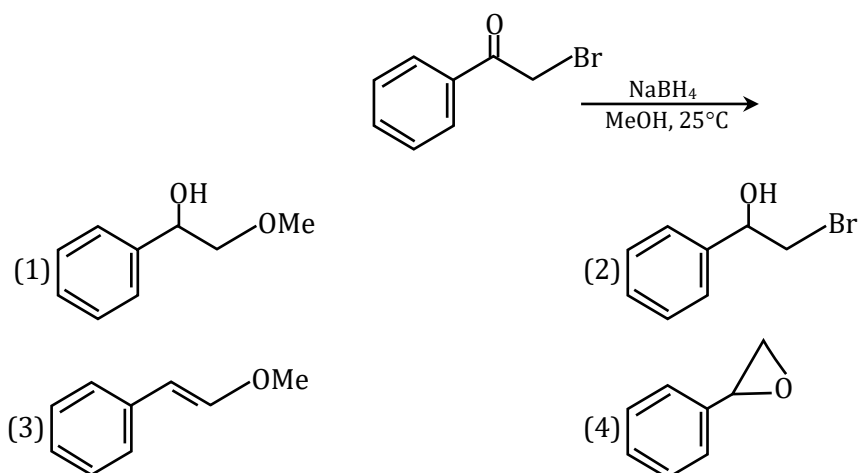
[JEE (Main) 2019]



CRD040

12. The major product of the following reaction is :

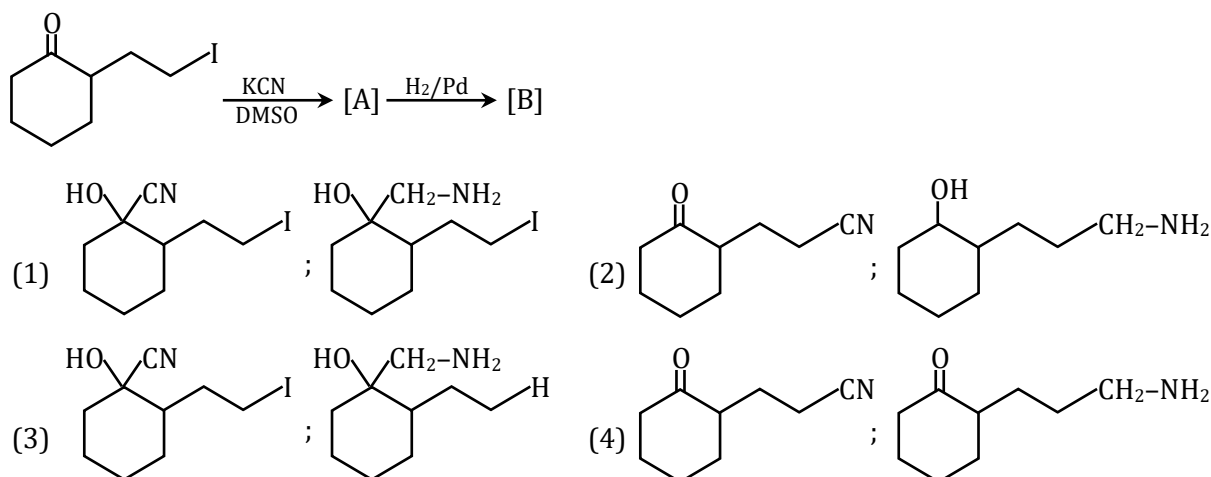
[JEE (Main) 2019]



CRD041

13. The major products A and B for the following reactions are, respectively:

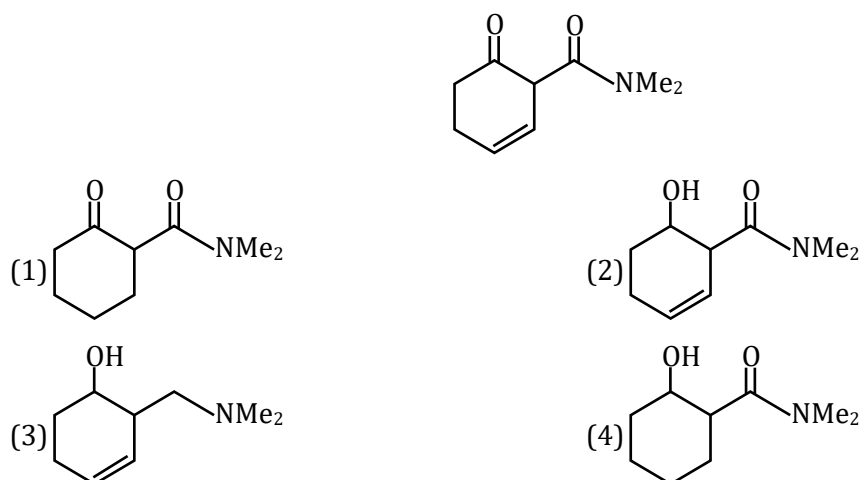
[JEE (Main) 2019]



CRD042

14. The main reduction product of the following compound with NaBH_4 in methanol is :-

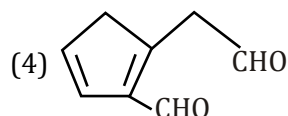
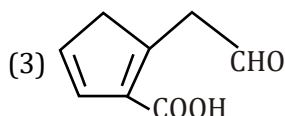
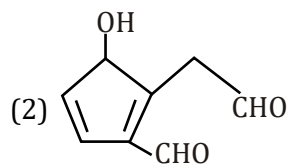
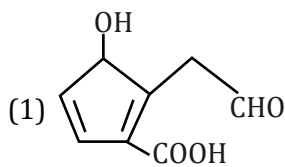
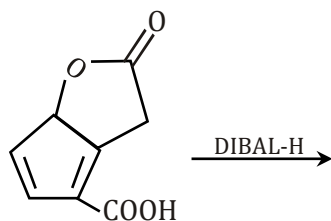
[JEE (Main) 2018]



CRD034

15. The major product obtained in the following reaction is :

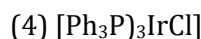
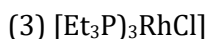
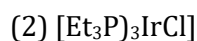
[JEE (Main) 2017]



CRD031

16. Wilkinson catalyst is :

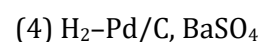
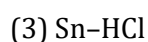
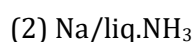
[JEE (Main) 2019]



CRD036

17. The trans-alkenes are formed by the reduction of alkynes with :

[JEE (Main) 2018]



CRD033

18. Given below are two statements :

[JEE (Main) 2023]

Statement I : under Clemmensen reduction conditions will give

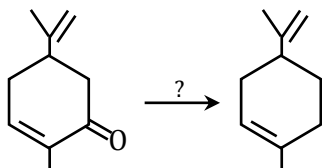
Statement II : under Wolff-Kishner reduction condition will give

In the light of the above statements, choose the correct answer from the options given below :

- (1) Statement I is false, but Statement II is true
- (2) Both Statement I and Statement II are false
- (3) Statement I is true, but Statement II is false
- (4) Both Statement I and Statement II are true

CRD050

19.



[JEE (Main) 2021]

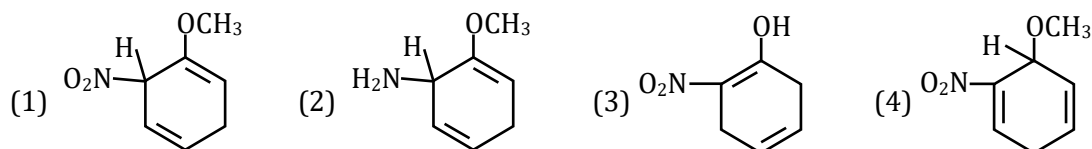
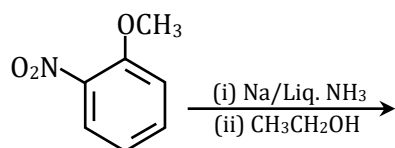
Which of the following reagent is suitable for the preparation of the product in the above reaction?

- (1) NaBH_4 (2) $\text{NH}_2\text{-NH}_2/\text{C}_2\text{H}_5\text{O}^\ominus\text{Na}^\oplus$
 (3) Ni/H_2 (4) $\text{Red P} + \text{Cl}_2$

CRD046

20. The major product of the following reaction is

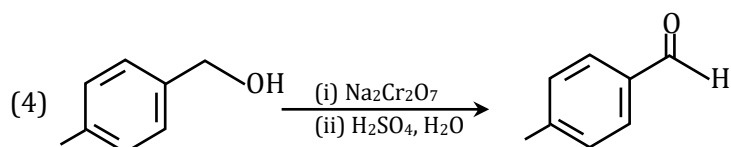
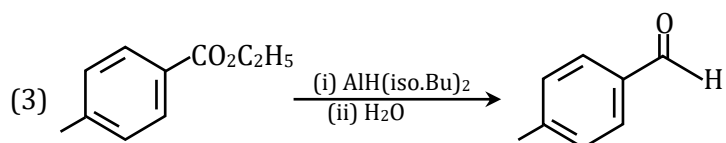
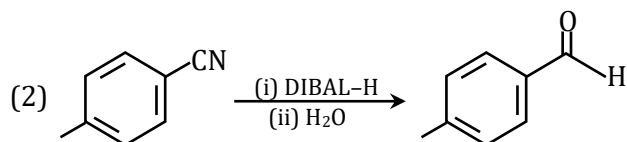
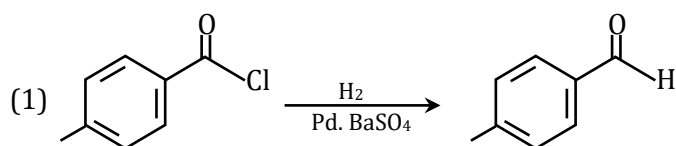
[JEE (Main) 2022]



CRD047

21. Which one of the following reactions does not represent correct combination of substrate and product under the given conditions ?

[JEE (Main) 2022]



CRD048

22. Glucose on prolonged heating with HI gives :

[JEE (Main) 2018]

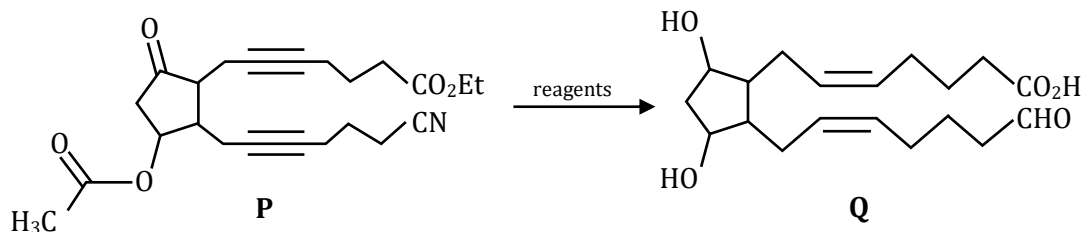
- (1) 1-Hexene (2) Hexanoic acid (3) 6-iodohexanal (4) n-Hexane

CRD032

EXERCISE - JEE (Advanced) PYQ

1. The option(s) with correct sequence of reagents for the conversion of **P** to **Q** is(are) :

[JEE (Advanced) 2024]



- (A) i) Lindlar's catalyst, H_2 ; ii) $SnCl_2/HCl$; iii) $NaBH_4$; iv) H_3O^+
 (B) i) Lindlar's catalyst, H_2 ; ii) H_3O^+ ; iii) $SnCl_2/HCl$; iv) $NaBH_4$
 (C) i) $NaBH_4$; ii) $SnCl_2/HCl$; iii) H_3O^+ ; iv) Lindlar's catalyst, H_2
 (D) i) Lindlar's catalyst, H_2 ; ii) $NaBH_4$; iii) $SnCl_2/HCl$; iv) H_3O^+

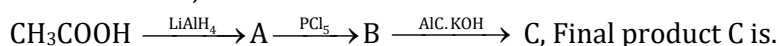
CRD134

2. Consider all possible isomeric ketones including stereoisomers of MW = 100, All these isomers are independently reacted with $NaBH_4$ (**NOTE** : stereoisomers are also reacted separately). The total number of ketones that give a racemic product(s) is/are.

[JEE (Advanced) 2014]

CRD081

3. In the reaction,



[JEE (Advanced) 2014]

- (A) Ethylene (B) Acetyl chloride (C) Acetaldehyde (D) Acetylene

CRD082

4. When $CH_2=CH-COOH$ is reduced with $LiAlH_4$, the compound obtained will be -

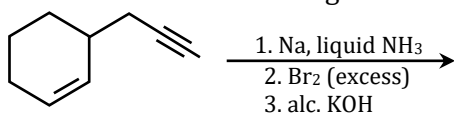
[AIEEE 2003]

- (A) $CH_3-CH_2-CH_2OH$ (B) CH_3-CH_2-CHO (C) CH_3-CH_2-COOH (D) $CH_2=CH-CH_2OH$

CRD076

5. The number of isomeric tetraenes (NOT containing sp -hybridized carbon atoms) that can be formed from the following reaction sequence is _____.

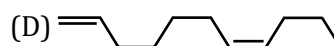
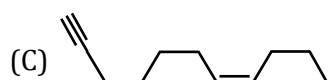
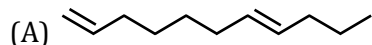
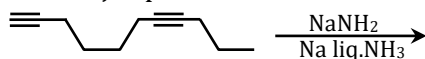
[JEE (Advanced) 2022]



CRD085

6. The major product formed in the following reaction is

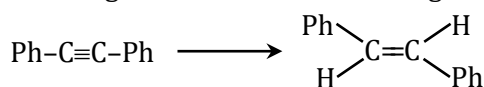
[JEE (Advanced) 2021]



CRD084

7. The reagent needed for converting is :

[JEE (Advanced) 2014]



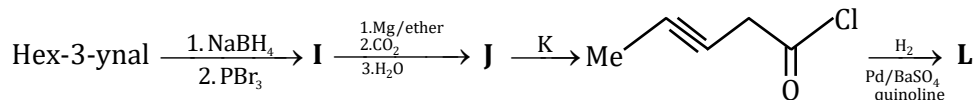
- (A) H_2 / Lindlar Cat.
 (C) $LiAlH_4$

- (B) Cat. hydrogenation
 (D) Li / NH_3

CRD083

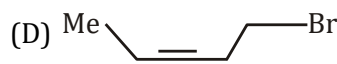
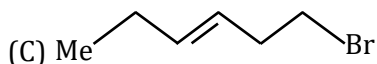
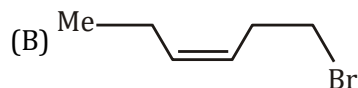
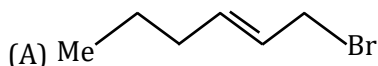
Paragraph for Question No. 08 to 10

In the following reaction sequence, products **I**, **J** and **L** are formed. **K** represents a reagent.



8. The structure of the product **I** is

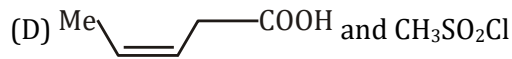
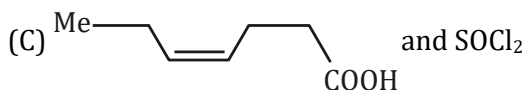
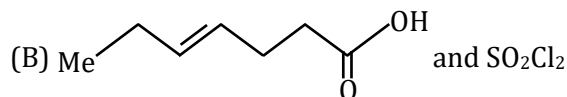
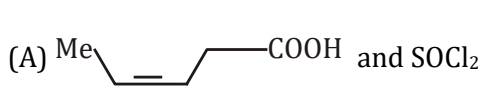
[JEE (Advanced) 2008]



CRD078

9. The structures of compounds **J** and **K**, respectively, are

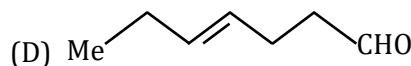
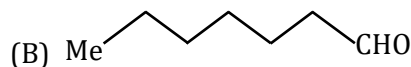
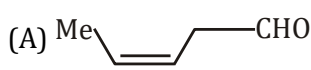
[JEE (Advanced) 2008]



CRD079

10. The structure of product **L** is :

[JEE (Advanced) 2008]



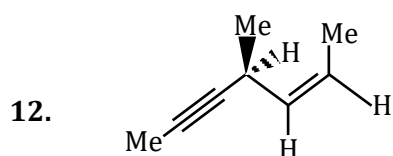
CRD080

11. 2-hexyne can be converted into trans-2-hexene by the action of :

[IIT 2004]

- (A) H_2 -Pd-BaSO₄ (B) Li in liq. NH_3 (C) H_2 -PtO₂ (D) NaBH_4

CRD077



Hydrogenation of the above compound in the presence of poisoned palladium catalyst gives-

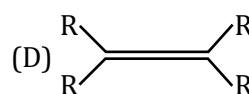
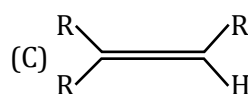
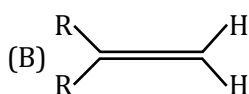
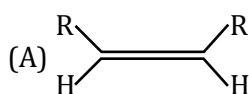
[IIT 2001]

- (A) An optically active compound (B) An optically inactive compound
(C) A racemic mixture (D) A diastereomeric mixture

CRD075

13. Which one of the following alkenes will react fastest with H_2 under catalytic hydrogenation condition

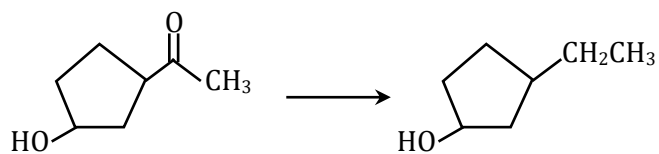
[IIT 2000]



CRD073

14. The appropriate reagent for the following transformation:

[IIT 2000]



- (A) Zn (Hg), HCl (B) NH_2NH_2 , $\text{OH}^\ominus$ (C) H_2/Ni (D) NaBH_4

CRD074

15. Under Wolff Kishner reduction conditions, the conversions which may be brought about is?

[IIT 1995]

- (A) Benzaldehyde into Benzyl alcohol (B) Cyclohexanol into Cyclohexane
(C) Cyclohexanone into Cyclohexanol (D) Benzophenone into Diphenylmethane

CRD072

16. Hydrogenation of benzoyl chloride in the presence of Pd and BaSO_4 gives :

[IIT 1992]

- (A) Benzyl alcohol (B) Benzaldehyde (C) Benzoic acid (D) Phenol

CRD071

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	A	D	B	A	C	D	D	A

EXERCISE (O-2)

Que.	1	2	3	4	5	6	
Ans.	ABCD	ABC	ABC	ABC	ABD	C	

EXERCISE (S)

Que.	1	2	3	4	5	
Ans.	5	3	3	1.06	4.00	

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	1	3	1	3	3	4	4	2	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	4	2	2	1	1	2	3	2	1
Que.	21	22								
Ans.	4	4								

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	CD	5	A	D	2	B	D	D	A	C
Que.	11	12	13	14	15	16				
Ans.	B	B	A	B	D	B				